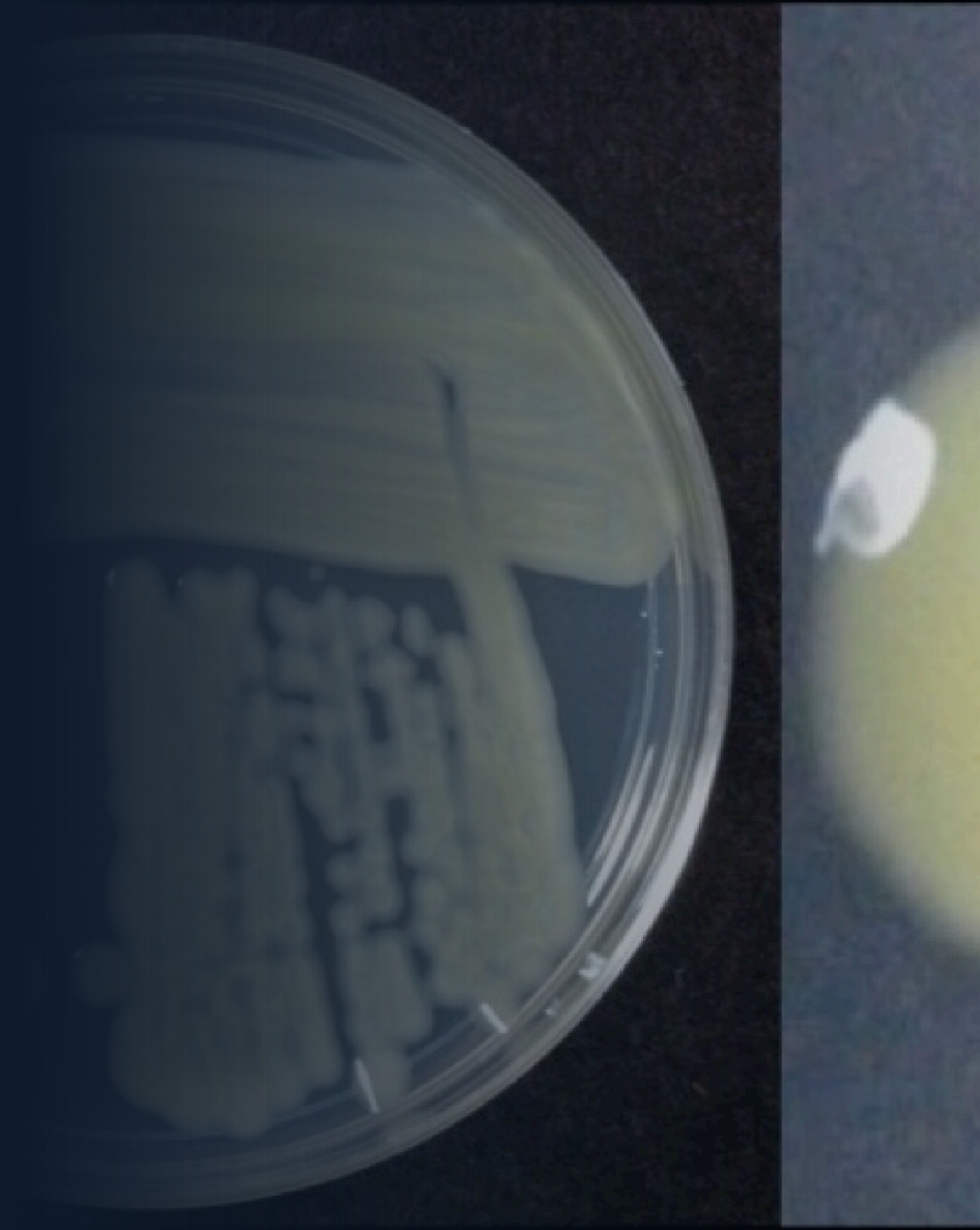
WORLD SCIENCE ACADEMY

SAMPLE LESSON · CASE STUDY: THE MICROBIAL WORLD

How Do You Study Something You Can't See?

A look inside a World Science Academy research session — where the invisible world of microbes becomes award-winning research.

Microbes · Disease · Data · 01



TODAY'S CHALLENGE

Microbes are everywhere — and mostly invisible.

They keep us alive, and they make us sick. Can we grow, see, and control them?

A microscopic fungus leaves visible marks — leaf-blight lesions on a sampled crop.

BY THE END, STUDENTS CAN

- Culture and identify a microbe from a real sample
- Read real inhibition & growth data to draw conclusions
- Explain how microbes evolve resistance over time

WHY MICROBES MATTER

The smallest things run the planet

Bacteria, fungi, and viruses are far too small to see — yet they shape our food, our health, and every ecosystem on Earth.

Everywhere

Trillions, unseen

A single gram of soil can hold billions of microbes.

Essential

Mostly helpful

They make our food, guard our gut, and recycle nutrients.

Dangerous

Some cause disease

Pathogens spread fast — and evolve to resist our defences.

HOW WE STUDY A MICROBE

Four steps to see the invisible

1 · Sample
Collect from a plant, surface, or swab.

2 · Culture
Grow it on agar until colonies appear.

3 · Observe
Identify it under the microscope.

4 · Test
Measure what slows or stops its growth.

You can't see one microbe — but a billion of them growing together, you can measure.

KEY CONCEPTS

The language students build

Microbe
A living thing too small to see — bacteria, fungi, and more.

Pathogen
A microbe that causes disease in a host.

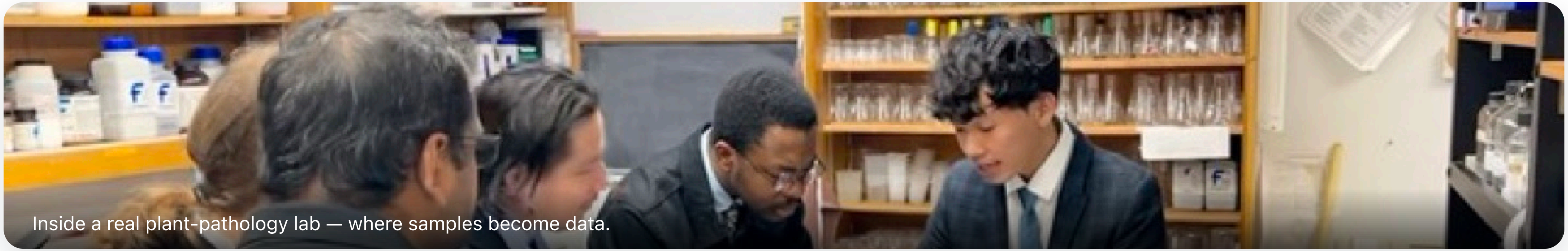
Culture & agar
Growing microbes on a nutrient gel so we can count and study them.

Resistance
When a microbe evolves to survive what used to kill it.

Under the microscope: the spores of a real plant pathogen.

HOW THIS SESSION RUNS

Students do, not just listen


Inside a real plant-pathology lab — where samples become data.

In-Person

- Real-time polling & word clouds
- Think-Pair-Share discussions
- Live dataset & graph analysis
- Case-study debates

Online

- Social annotation of case studies
- Micro-module self-checks
- Small-group discussion pods
- Microscopy & image analysis

READING REAL DATA

What actually stops a microbe?

Students place each treatment on a bacterial lawn and measure the clear **zone of inhibition** — bigger means stronger.

Treatment	Zone of inhibition	Effect
Household disinfectant	**22 mm**	**Strong**
Alcohol hand sanitizer	15 mm	Good
Tea-tree oil	9 mm	Mild
Garlic extract	6 mm	Weak
Water (control)	**0 mm**	**None**

Illustrative teaching figures — the point is how to measure and compare, not exact millimetres.

THE BIG IDEA

Killing microbes isn't always the goal

Most microbes help us — wiping them all out would do more harm than good. And the more we use a treatment, the faster microbes **evolve resistance** to it. The real skill is measuring carefully and **using the right tool at the right time.**

Good microbes vs. bad microbes

Overuse breeds resistance

Measure before you conclude

350 YEARS OF DISCOVERY

How we learned to see the unseen

- **1676** — **Leeuwenhoek** sees the first "animalcules" through a homemade lens.
- **1860s** — **Pasteur** shows microbes cause spoilage & disease — germ theory.
- **1884** — **Koch's postulates** — how to prove a microbe causes a disease.
- **1928** — **Fleming** discovers penicillin — the antibiotic era begins.
- **1940s+** — Microbes fight back — **antibiotic resistance** spreads.
- **Today** — **Genomics & rapid ID** — we read a microbe's DNA in hours.

The same core skill, 350 years on: culturing microbes on a plate to study them.

FROM QUESTION TO PROJECT

Turning this into a science-fair study

Ask a testable question

- Which everyday substance best stops bacteria?
- How does temperature change microbial growth?
- Do microbes regrow after a treatment — and how fast?

Design a fair test

- Controls, replicates & sterile technique
- Measure objectively (mm, colony counts)
- Graph the data — then argue from it

Real research means collecting real data — scoring disease in the field, sample by sample.

Our role: we coach students through each step — framing the question, running the experiment safely, analysing results, and presenting like a real researcher.

WHAT STUDENTS WALK AWAY WITH

Deliverables

01

A research question

A focused, testable hypothesis they can defend.

02

Experiment & data

A clean protocol, real measurements, and graphs.

03

Poster & talk

Competition-ready board and a confident presentation.

Scientific writing

An abstract and report in real academic structure.

Judge-ready Q&A

Practice fielding tough questions like a researcher.

Presenting findings with confidence.

MICROBIOLOGY & LIFE SCIENCE · SAMPLE LESSON

From the invisible world to a research question

This is one of many topics your child can explore with us — real samples, real data, and a project worthy of the world stage.

World Science Academy

Helping Your Child Take the World Stage